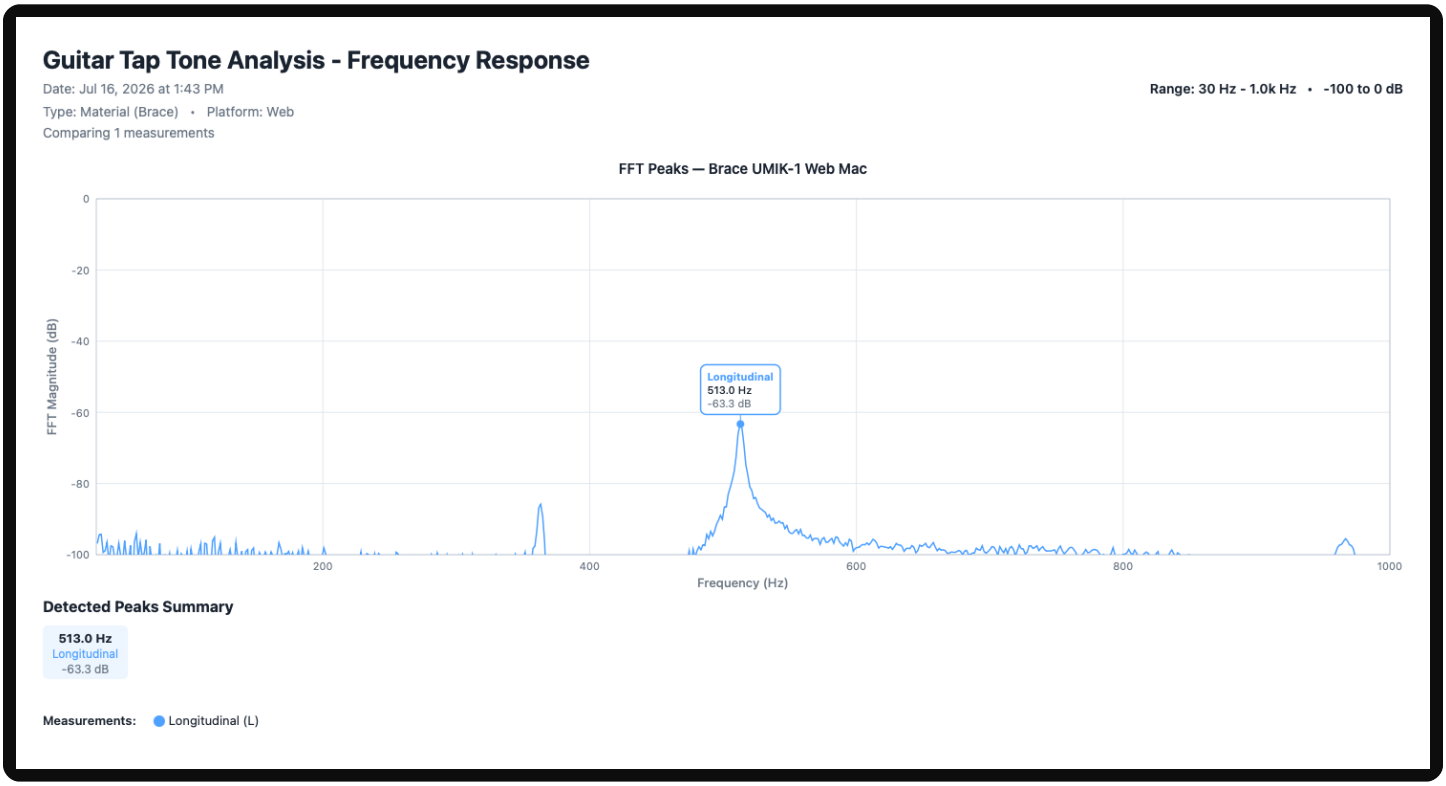
# GuitarTap

Tap Tone Analysis Report

Jul 16, 2026 at 1:43 PM

**Measurement Name:** Brace UMIK-1 Web Mac
**Type:** Material (Brace)
**Frequency Range:** 30.0 Hz – 1.0 kHz
**Microphone:** Umik-1  Gain: 18dB · calibrated (7108913)

## Frequency Spectrum




## Detected Peaks

| Frequency | Magnitude | Note | Q Factor | Role |
|---|---|---|---|---|
| 513.0 Hz | -63.3 dB | C5 | 87.5 | fL (Longitudinal) |

## Brace Properties

### Sample Dimensions

Length: **556.0 mm**
Width: **20.4 mm**
Thickness: **29.35 mm**
Mass: **128.0 g**
Density: **0.385 g/cm³**

fL: **513.0 Hz**

Speed of Sound: **5257 m/s**
Specific Modulus: **27.6** (Excellent)
Young's Modulus (E): **10.64 GPa**
Radiation Ratio: **13.7**

**Overall Quality:** **Excellent**

**Single-Tap Measurement (fL only):**

- **1. Longitudinal (fL) Tap**
  Hold brace at 22% from one end along the length. Tap center.

*The strongest peak is auto-selected.*